FULL NAME: ____________________ **BLOCK:** __________

Reciprocal Function + Translations — Assessment-Critical

Lesson 4-1 · Period 3 · Student Packet

OBJECTIVES

Math Objective	Graph the reciprocal function $f(x) = 1/x$ and its translations $g(x) = 1/(x-h)+k$. Identify vertical and horizontal asymptotes, state domain and range, and find intercepts where they exist.
Language Objective	Students will use the frame "This graph is the parent $f(x) = 1/x$ translated ___ units ___. The asymptotes are at $x =$ ___ and $y =$ ___." to describe any translated reciprocal function.
Essential Understanding	The reciprocal function has a vertical asymptote at $x = 0$ and horizontal asymptote at $y = 0$. Translations shift the asymptotes: $g(x) = 1/(x-h)+k$ has asymptotes at $x = h$ and $y = k$. The point (h, k) is the new "center."

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students will identify asymptotes and describe translations for 6 reciprocal-function items. This is Topic 4 assessment-prep material.
Essential Question	How do the values of h and k in $g(x) = 1/(x-h)+k$ determine the graph's asymptotes, domain, range, and intercepts?
Materials	Pencils; student packet; graphing calculator (optional, for verification).

LAUNCH — Reciprocal Function + Translations (teacher models Ex 4 + Ex 5)

[BOOK] RECIPROCAL FUNCTION — PARENT $f(x) = 1/x$

- Vertical asymptote: $x = 0$
- Horizontal asymptote: $y = 0$
- Domain: $\{x \mid x \neq 0\}$ • Range: $\{y \mid y \neq 0\}$
- Graph: hyperbola in Quadrants I and III (two branches, no intersection)

[BOOK] TRANSLATED RECIPROCAL — $g(x) = 1/(x-h)+k$

- Vertical asymptote shifts to: $x = h$
- Horizontal asymptote shifts to: $y = k$
- Domain: $\{x \mid x \neq h\}$ • Range: $\{y \mid y \neq k\}$
- The point (h, k) is the "new center" — where the asymptotes cross.
- Example: $g(x) = 1/(x-3)+2 \to$ asymptotes $x = 3$ and $y = 2$.

To find INTERCEPTS:

- y-intercept: set $x = 0$. $y = 1/(0-h)+k = -1/h+k$.
- x-intercept: set $y = 0$. Solve $0 = 1/(x-h)+k \to x = h - 1/k$.

EXPLORE — Think-Pair-Share (35 min)

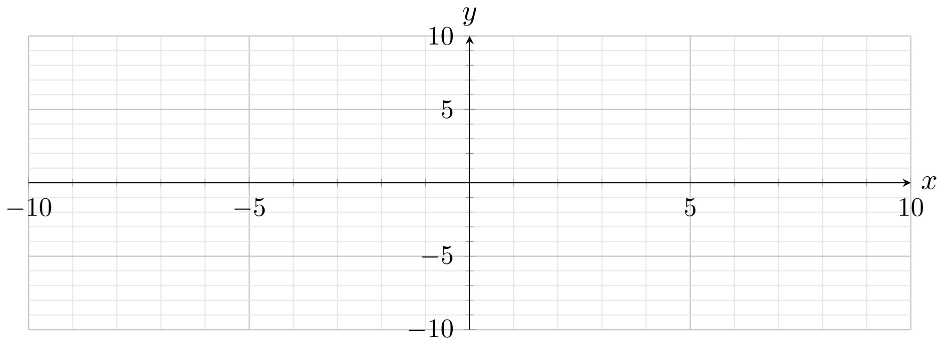
Work with a partner. Practice #19 is assessment-critical (asymptotes + intercepts) — plan ∼8 min for it.

Bank-item labels (in order):

1. Try It 4 — Graph $g(x) = 10/x$ (scaled parent)
2. Practice #11 — Error Analysis: $y = 5/x$
3. Practice #18 — Graph $y = -2/x$ (reflected parent)
4. Try It 5 — Graph $g(x) = 1/(x + 2) - 4$ (translated)
5. Practice #19 [STAR] ASSESSMENT CRITICAL — $g(x) = 1/(x - 2) + 6$ + intercepts
6. Practice #21 — Video game storage (modeled)

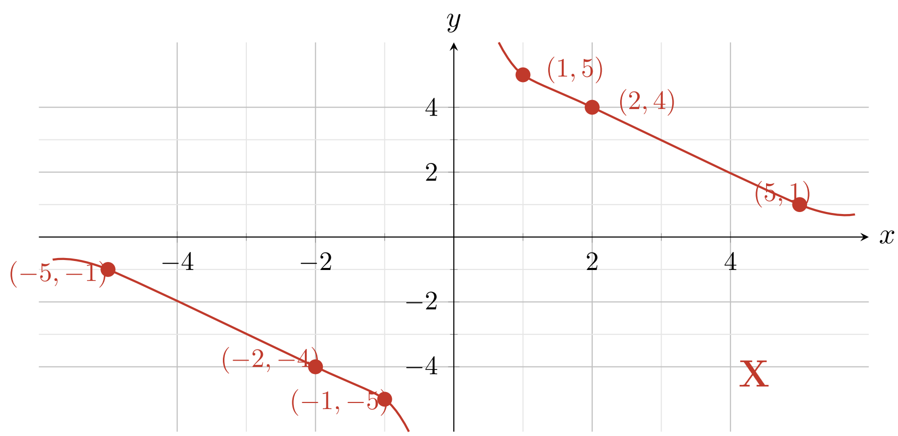
Try It 4 — Graph $g(x) = 10/x$ (scaled parent)

Try It 4. Graph the function $g(x) = 10/x$. What are the domain, range, and asymptotes of the function?



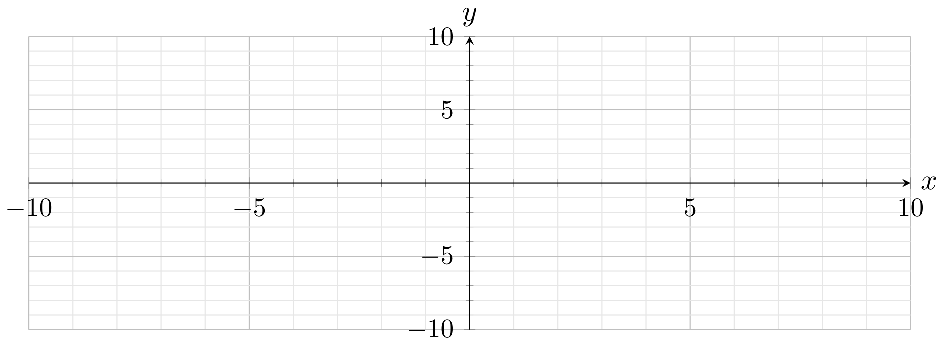
Practice #11 — Error Analysis: $y = 5/x$

11. Error Analysis: Describe and correct the error a student made in graphing the function $y = 5/x$. The student's graph shows labeled points $(1, 5)$, $(2, 4)$, $(5, 1)$ in Quadrant I and $(-5, -1)$, $(-2, -4)$, $(-1, -5)$ in Quadrant III, with a large red X indicating the graph is incorrect.



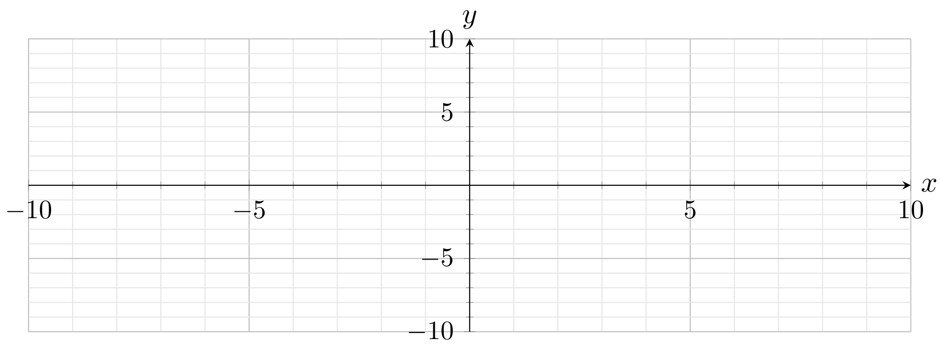
Practice #18 — Graph $y = -2/x$ (reflected parent)

18. Graph the function $y = -2/x$. What are the domain, range, and asymptotes of the function?



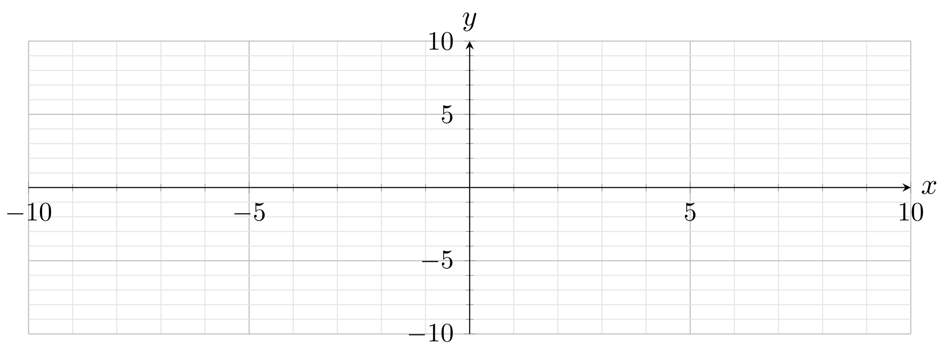
Try It 5 — Graph $g(x) = 1/(x+2) - 4$ (translated)

Try It 5. Graph $g(x) = 1/(x+2) - 4$. What are the equations of the asymptotes? What are the domain and range?



Practice #19 [STAR] ASSESSMENT CRITICAL — $g(x) = 1/(x-2) + 6$ + intercepts

19. Graph $g(x) = 1/(x-2) + 6$. What are the equations of the asymptotes? What are the domain and range? What are the intercepts?



Practice #21 — Video game storage (modeled)

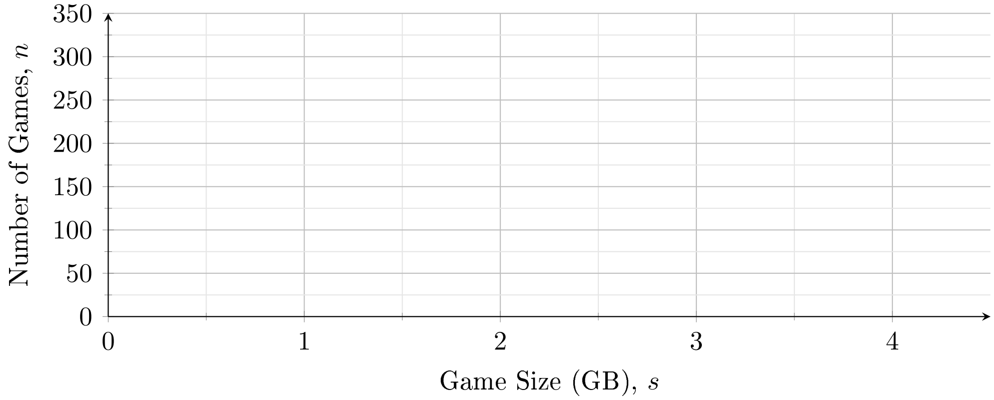
21. Use Structure: The number of downloaded games that can be stored on a video game system varies inversely with the average size of a video game. A certain video game system can store 160 games when the average size of a game is 2.0 gigabytes (GB).

(a) Write an inverse equation that relates the number of games n that will fit on the video game system as a function of the average game size s in GB.

(b) Use the inverse relationship to complete the table of values.

Game Size (GB), s	1.0	2.5	3.0	4.0
Number of Games, n	$\square$	$\square$	$\square$	$\square$

(c) Sketch a graph of this inverse relationship on a coordinate plane.



SHARE / SUMMARY (5 min)

Sentence frames
Pair share (#19): "This graph is the parent $f(x) = 1/x$ translated ___ units ___ and ___ units ___. The asymptotes are $x =$ ___ and $y =$ ___. The intercepts are ___."
Whole class: one pair reads #19. Class signals thumbs up/sideways.

[CHART] CONCEPT SUMMARY (your reference for Topic 4 assessment)

INVERSE VARIATION ($y = k/x$):
— WORDS: As one variable increases, the other decreases; their product k is constant.
— ALGEBRA: $y = k/x$ where $k \neq 0$
— GRAPH: parent hyperbola, asymptotes $x = 0$ and $y = 0$

RECIPROCAL FUNCTION TRANSLATIONS ($y = a/(x - h) + k$):
— WORDS: The reciprocal function can shift left/right (h) and up/down (k).
— ALGEBRA: $y = a/(x - h) + k$
— GRAPH: asymptotes shift to $x = h$ and $y = k$; new "center" at (h, k).

EXIT

I can identify asymptotes of $y = 1/(x - h) + k$. Confidence: 1 / 2 / 3 / 4 / 5
I can describe a translation from $f(x) = 1/x$ to $g(x)$. Confidence: 1 / 2 / 3 / 4 / 5
One thing I want to review before Topic 4 assessment: ....

OPTIONAL REINFORCEMENT (not collected)

Extra stretch for fast finishers. Try without a calculator first.

Optional #1 (Savvas SAT/ACT style)

25. SAT/ACT. In a given set of experiments, t and w always vary inversely. When $t = 0.4$ seconds, $w = 172.8$ grams. How much time has passed when $w = 14.4$ grams?

A. 2.4 s

B. 4.8 s

C. 27.648 s

D. 995.328 s